



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Information Technology

Academic Year: 2024 - 2025(Even Semester)

Degree, Semester & Branch: B. Tech, IV & IT

Course Code & Title: CS3492 & Database Management Systems

Name of the Faculty member: A. Alagulakshmi, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit IV/ Algorithms for Selection, Sorting and join operations
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO 11
- **Activity Chosen:** Collaborative Coding
- **Justification:**

Collaborative Coding is a student-centered teaching strategy where learners work together to solve problems and build understanding. Students explored essential query processing algorithms including selection, sorting, and join operations, which are critical for optimizing performance in DBMS. Through collaborative coding, students applied these algorithms to real-world scenarios, enhancing both their technical and analytical skills. Team-based work encouraged discussion on algorithm selection, implementation, and performance trade-offs. The activity reinforced practical applications and fostered peer-to-peer knowledge sharing.

- **Time Allotted for the Activity:** 20 Minutes

- **Details of the Implementation:**

After completing Unit IV, students were divided into groups to collaboratively explore and simulate key algorithms used in query processing, including selection, sorting, and join operations. For **selection operations**, students analyzed queries with various predicates to understand how different conditions affect query outcomes and performance. In the **sorting techniques** section, they examined and compared multiple sorting algorithms, including external sorting methods commonly used in handling large datasets. When studying **join algorithms**, students explored different join strategies such

as nested loop join, sort-merge join, and hash join, focusing on their operational mechanisms, benefits, and the scenarios in which each is most effective. Finally, each group engaged in **discussion and presentation**, where they justified their choice of algorithms for specific query plans based on efficiency and suitability, enhancing both their analytical thinking and presentation skills.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO1
CO4	1	2	3	2	3	2	3	3

- **PO / PSO mapped:**

Innovative Practice	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO1
	1	2	3	2	3	2	3	3
Justification for correlation	Apply foundational and domain knowledge to address technical challenges.	Identify and analyze engineering problems, formulating relevant solutions.	Design effective and optimized solutions for complex database queries.	Conduct investigations and simulations to evaluate algorithm performance.	Collaborate effectively within teams, promoting shared problem-solving.	Communicate ideas clearly through presentations and discussions.	Engage in lifelong learning by exploring evolving technologies in DBMS.	Apply key database concepts to build and optimize practical DBMS solutions.

(1 – Low 2 – Moderate 3 – High)

- **Images / Screenshot of the practice:**



Figure 1: Student discussed with their team



Figure 2: Student Share their ideas

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students expressed satisfaction in learning collaboratively.
- The group activity promoted active engagement and motivated students to share their ideas confidently.

- ❖ ***Benefit of the practice:***

- Every student actively participated in discussions and problem-solving exercises.
- Teamwork enhanced both conceptual understanding and communication skills.
- The activity strengthened students' grasp of selection, sorting, and join algorithms in query processing.
- It improved students' ability to analyze algorithm performance and choose suitable techniques for different database scenarios.
- Encouraged logical reasoning and helped connect theory with real-world applications.

- ❖ ***Challenges faced in implementation:***

- In some groups, high-performing students dominated discussions, limiting equal contribution.
- A few students struggled to clearly articulate their understanding of algorithm efficiency and internal DBMS operations.
- Limited time restricted deeper exploration of more complex join strategies.

References:

- ❖ <https://www.valamis.com/hub/collaborative-learning>
- ❖ <https://teaching.cornell.edu/teaching-resources/engaging-students/collaborative-learning>
- ❖ <https://www.evergreen.edu/sites/default/files/facultydevelopment/docs/WhatIsCollaborativeLearning.pdf>

Signature of Faculty Member

HOD

